# Electrical Circuits II

## (EE-1201)

Lecture-04

**Network Theorems**

**(Norton's Theorem)**

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# Outlines

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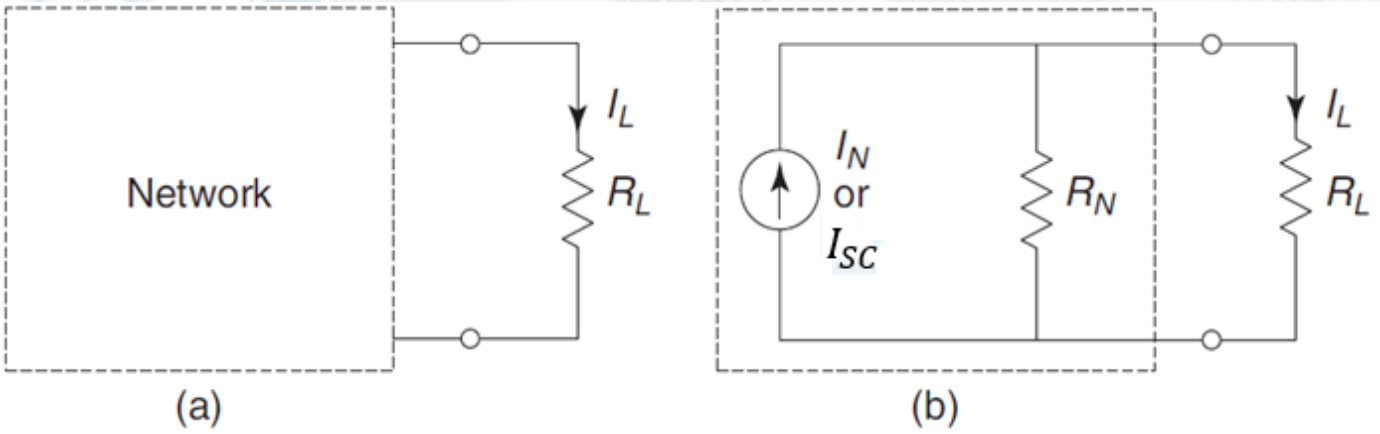
**This unit covers:**

- Norton's theorem
- Numerical examples

# Norton's Theorem

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- It states that *"any two terminals of a network can be replaced by an equivalent current source and an equivalent parallel resistance. The constant current is equal to the current which would flow in a short circuit placed across the terminals. The parallel resistance is the resistance of the network when viewed from these open-circuited terminals after all voltage and current sources have been removed and replaced by internal resistances."*

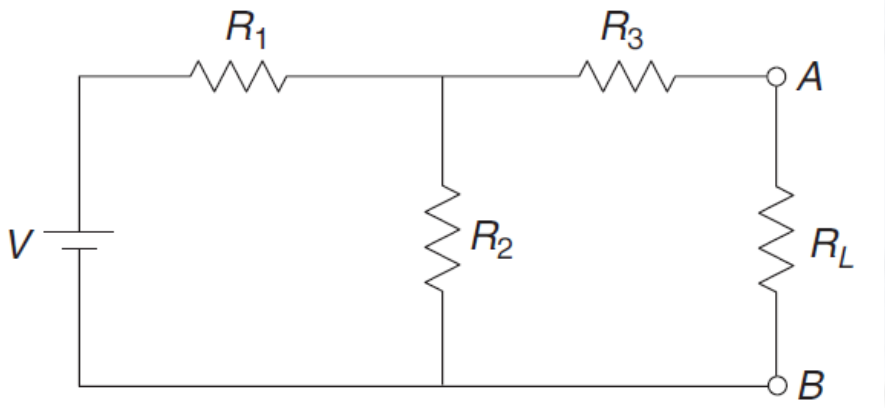



*Network illustrating Norton's theorem*

# Norton's Theorem

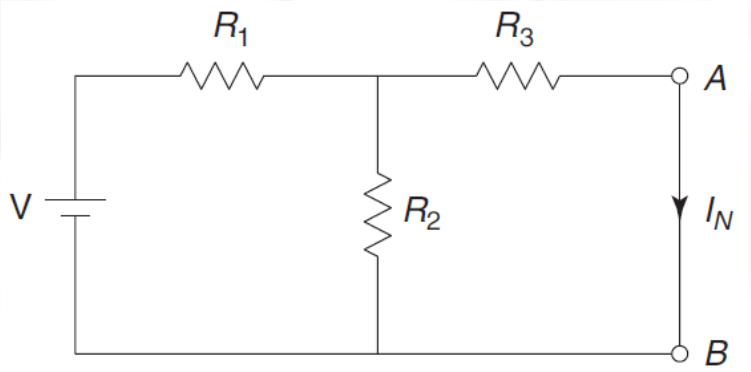
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**Explanation:** Consider a simple network as shown in Fig.




***Step 1: Norton's short circuit current $I_N$ calculation,***

- For finding load current through $\mathrm{R_L}$, first remove the load resistor $\mathrm{R_L}$ from the network and calculate short circuit current $I_{SC}$ or $I_N$ which would flow in a short circuit placed across terminals A and B as shown in Fig.
- *After simplifying compute current $I_N$*

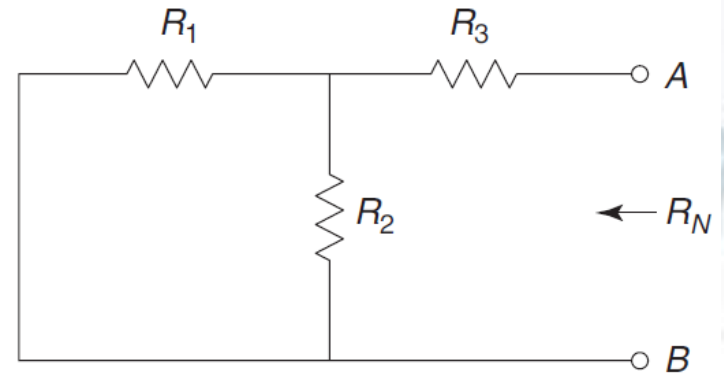


# Norton's Theorem

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## *Step 2: Norton's resistance $R_N$ calculation,*

- For finding parallel resistance $R_N$, replace the voltage source by a short circuit and calculate resistance between points A and B as shown in Fig

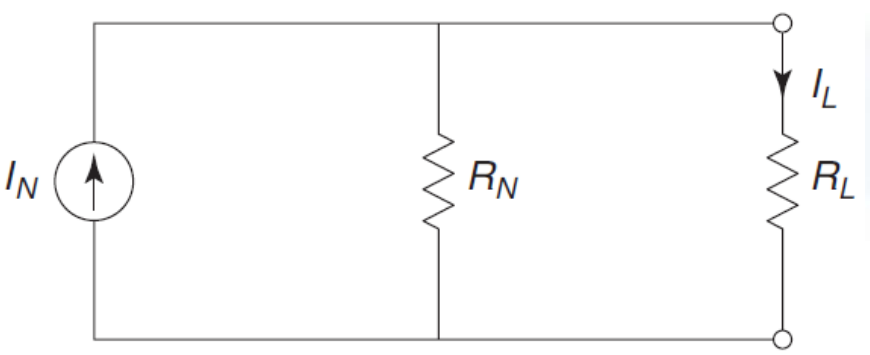



$$R_N = R_3 + \frac{R_1 R_2}{R_1 + R_2}$$

## *Step 3: Norton's equivalent network*

- Load current through $\mathrm{R_L}$ can be expressed as

$$I_L = I_N \frac{R_N}{R_N + R_L}$$



# Norton's Theorem

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***Network contains both independent and dependent sources***

- Norton's resistance $R_N$ is calculated as

$$R_N = \frac{V_{Th}}{I_N}$$

where $V_{Th}$ is the open-circuit voltage across the terminals A and B.

- Dependent sources are active at all times. They have zero values only when the control voltage or current is zero.
- $R_N$ may be negative in some cases which indicates negative resistance region of the device, i.e., as voltage increases, current decreases in the region and vice-versa.

# Norton's Theorem

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***Network contains only dependent sources***

- If the network contains only dependent sources then

$$V_{Th} = 0$$
$$I_N = 0$$

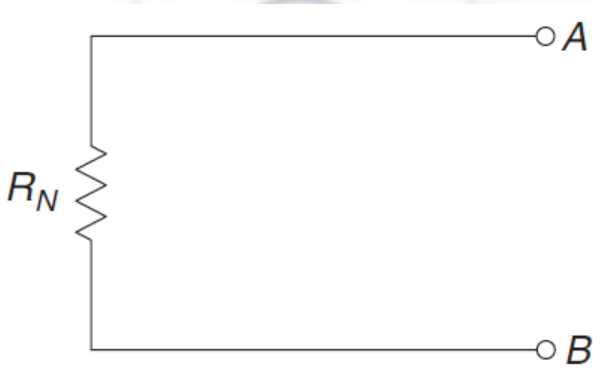
- For finding $R_N$ in such a network, a ***known voltage V is applied*** across the terminals A and B and ***current is calculated*** through the path AB

$$R_N = \frac{V}{I}$$

- *Alternatively*, a known ***current source I*** is connected across the terminals A and B and ***voltage is calculated*** across the terminals A and B

$$R_N = \frac{V}{I}$$

Norton's equivalent network for such a circuit is shown in the Fig



# Norton's Theorem

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## Steps to be Followed in Norton's Theorem

1. Remove the load resistance $R_L$ put a short circuit across the terminals
2. Find the short-circuit current $\mathrm{I}_{SC}$ or $I_N$ across A and B
3. Find the resistance $R_N$ as seen from points A and B
4. Replace the network by a current $I_N$ in parallel with resistance $R_N$
5. Find the current through $R_L$ using current division rule,

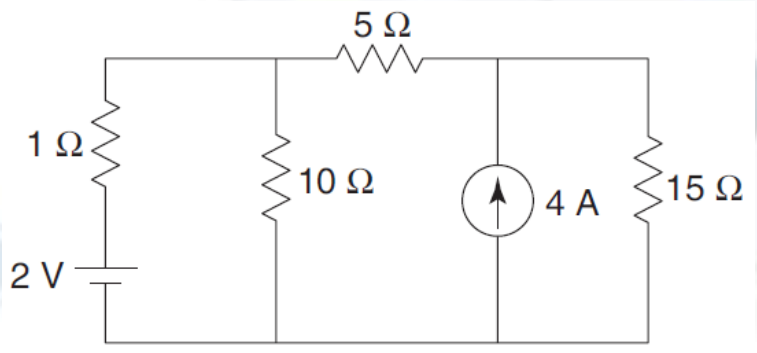
$$I_L = I_N \frac{R_N}{R_N + R_L}$$

# Norton's Theorem

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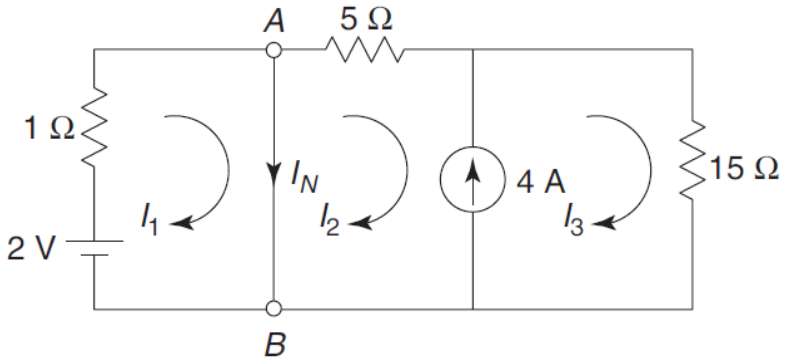
## Numerical Example-12:

Find current through 10 Ω resistor using Norton's equivalent in the given circuit.




## Solution:

Step I: Calculation of $I_N$,




- Applying KVL to Mesh 1;

$$2 - 1I_1 = 0$$

$$I_1 = 2\ A \ldots . (i)$$

- Meshes 2 and 3 will form a supermesh.
- Writing the current equation for the supermesh,

$$I_3 - I_2 = 4 \ \ldots . (ii)$$

# Norton's Theorem

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- Applying KVL to the supermesh,

$$-5I_2 - 15I_3 = 0 \ldots . (iii)$$

- Solving Eqs (i), (ii) and (iii),

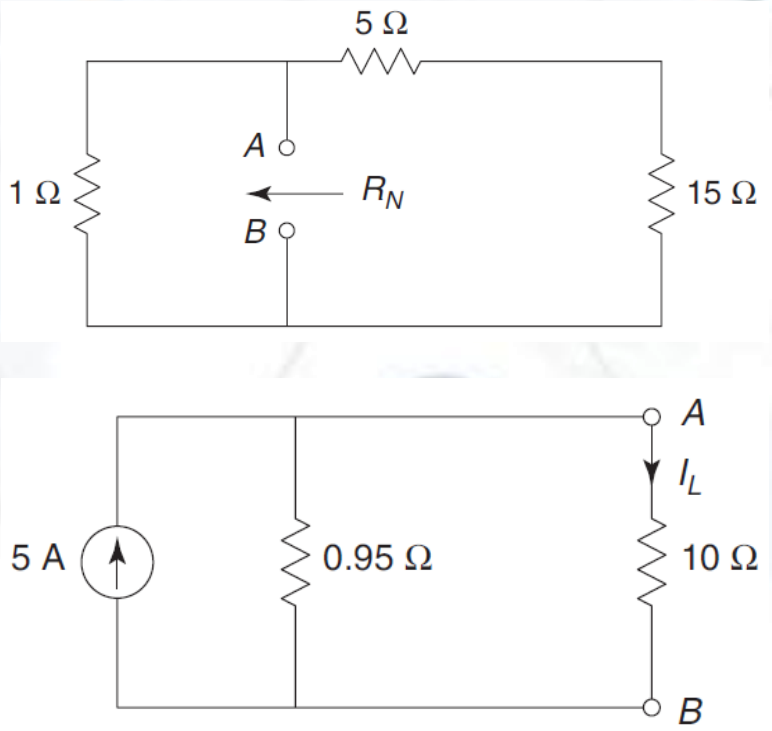
$$I_1 = 2\ A$$
$$I_2 = -3\ A$$
$$I_3 = 1\ A$$
$$I_N = I_1 - I_2 = 2 - (-3) = 5\ A$$

Step II: Calculation of $R_N$,

$$R_N = 1||(5 + 15) = 0.95\ \ \Omega$$

Step IV: Calculation of $I_L$,

$$I_L = I_N \frac{R_N}{R_N + R_L}$$
$$= 5 \frac{0.95}{0.95 + 10}$$
$$= 0.43\ A\ (\downarrow)$$

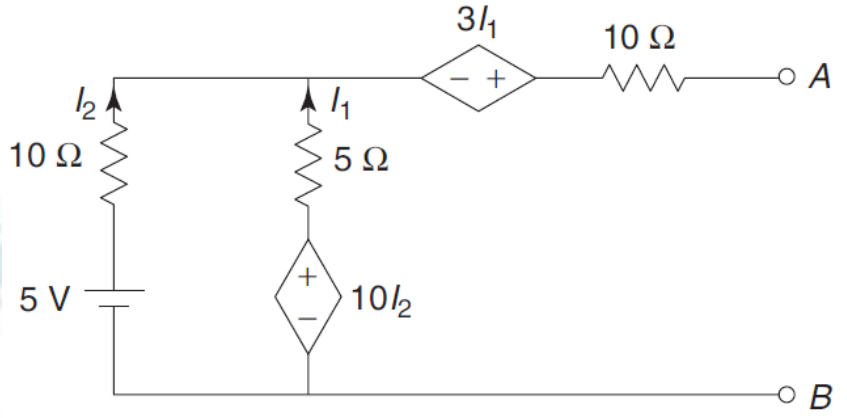


# Norton's Theorem

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## Numerical Example-13:

Find Norton's equivalent network across terminals A and B of Fig.



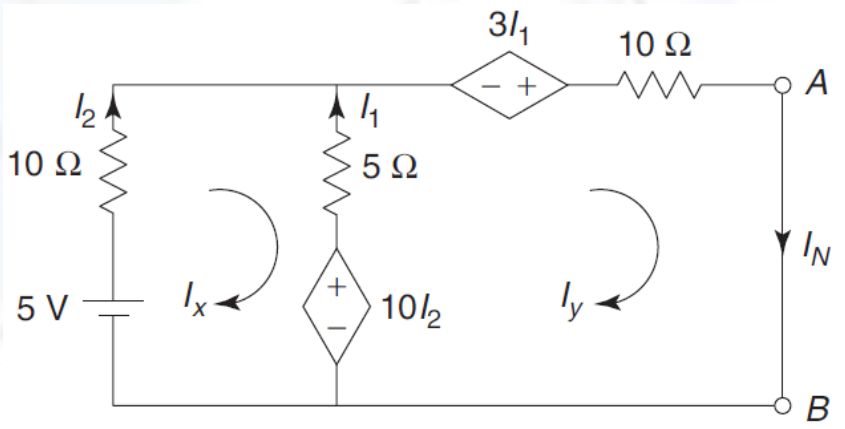

## Solution:

Step I: Calculation of $I_N$,

- From figure, it can be observed that;

$$I_2 = I_x \ldots . (i)$$

$$I_1 = I_y - I_x \ldots . (ii)$$



# Norton's Theorem

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- Applying KVL to Mesh 1;

$$5 - 10I_x - 5(I_x - I_y) - 10I_2 = 0$$

$$25I_x - 5I_y = 5 \ldots . (iii)$$

- Applying KVL to Mesh 2;

$$10I_2 - 5(I_y - I_x) + 3I_1 - 10I_y = 0$$

$$12I_x - 12I_y = 0 \ldots . (iv)$$

- Solving Eqs (iii) and (iv),

$$I_x = 0.25\ A$$

$$I_y = 0.25A$$

$$I_N = I_y = 0.25\ A$$

Step II: Calculation of $R_N$,

$$R_N = \frac{V_{Th}}{I_N}$$

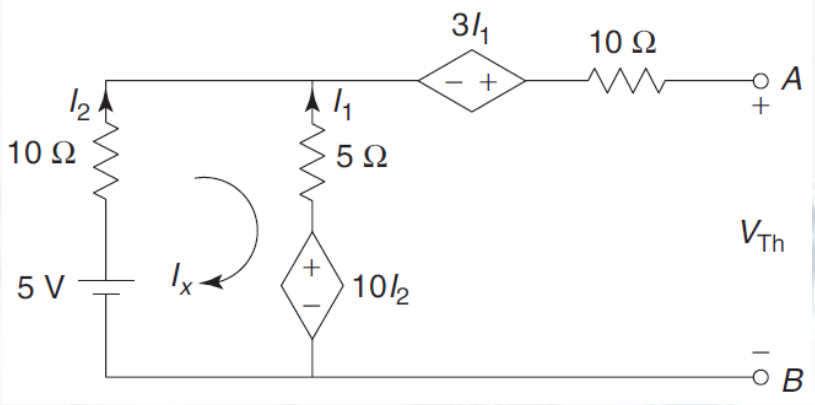
- Here, $V_{Th}$ is to be calculated

# Norton's Theorem

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- Calculation of $V_{Th}$,
- From figure, it can be observed that,

$$I_2 = I_x$$
$$I_1 = -I_x$$




- Applying KVL to the mesh,

$$5 - 10I_x - 5I_x - 10I_2 = 0$$
$$5 - 10I_x - 5I_x - 10I_x = 0$$
$$I_x = 0.2\ A$$
$$I_1 = -0.2\ A$$

- Writing the $V_{Th}$ equation,

$$5 - 10I_x + 3I_1 - V_{Th} = 0$$
$$5 - 10(0.2) + 3(-0.2) - V_{Th} = 0$$
$$V_{Th} = 2.4\ V$$

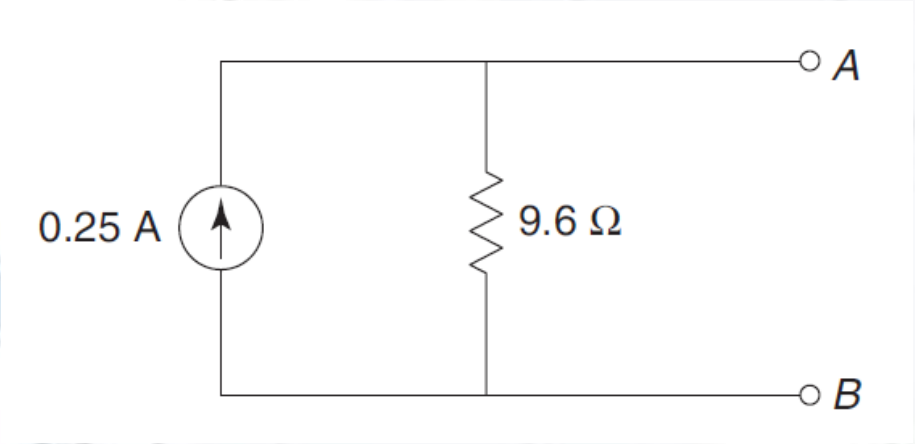
- Finally,

$$R_N = \frac{V_{Th}}{R_N} = \frac{2.4}{0.25} = 9.6\ \Omega$$

# Norton's Theorem

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Step III: Norton's Equivalent Network,



# Norton's Theorem

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## Homework #3:

Find the Norton's equivalent network across terminals a and b of Figure given below:

